MILITARY SPECIFICATION SHEET

PRINTED CIRCUIT CARD, M1 AND M1A1 TANK TURRET NETWORK BOX, TNB-A4

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Turret Network Box, TNB-A4 described herein shall consist of this specification and the latest use of MIL-P-62341.

3. REQUIREMENTS

3.4 Design and construction. The printed circuit card (PCC) shall contain relays which provide the control logic for the turret electrical system (see 4.8.1).

3.4.1 Configuration. The PCC shall be fabricated and assembled in accordance with Drawing 12282891 (see 4.8.2).

3.4.2 Weight. The weight of the PCC shall not exceed 1.5 pounds (lb) (see 4.8.3).

3.4.4 Nuclear hardening. The PCC shall withstand the nuclear threat environments as defined in classified documents control number DEOS 2060 (see 4.8.1).

3.5 Performance.

3.5.1 Circuit continuity. This characteristic is not specified.

3.5.2 Circuit isolation. The resistance between any two isolated circuits defined in table I shall be not less than 100 megohms at 100 ± 10 volts (V) direct current (de) (see 4.8.4.3).

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TABLE I. Isolated circuits.

Circuit Number	Terminals Common to the Respective Circuits.
1	P2-4, P2-8
2	P2-27, P2-28
3	P2-10, P2-12, P2-35
4	P1-4, P1-7
5	P1-32, P1-33, P1-34
6	P1-1, P1-2, P1-3, P1-5, P1-6, P1-8, P1-9, P1-10, P1-11, P1-12, P1-13, P1-14, P1-15, P1-16, P1-21, P1-22, P1-23, P1-24, P1-25, P1-26, P1-27, P1-28, P1-29, P1-30, P1-31, P1-35, P2-1, P2-2, P2-3, P2-6, P2-11, P2-13, P2-14, P2-15, P2-16, P2-17, P2-18, P2-19, P2-20, P2-21, P2-22, P2-23, P2-24, P2-25, P2-26, P2-29, P2-30, P2-32, P2-33, P2-34
7	P1-17 (unused)
8	P1-18 (unused)
9	P1-19 (unused)
10	P1-20 (unused)
11	P2-5 (unused)
12	P2-7 (unused)
13	P2-9 (unused)
14	P2-31 (unused)

3.5.3 Diode circuits.

3.5.3.1 Forward voltage drop. The forward voltage drop of the diode circuits specified in table 11 shall not exceed 1.3 V dc at 500 ± 50 milliamperes (mA) (see 4.8.4.4.1).

3.5.3.2 Reverse leakage current. The reverse leakage current of the diode circuits specified in table II shall not exceed 50 mA at 100 ± 10 V dc (see 4.8.4.4.2).

TABLE II. Diode circuits.

Diode Circuit	Terminals		18 to 30 V dc		Remarks
	Anode	Cathode	Positive	Return	
CR4	P2-29	P1-1			
CR5	P2-30	P1-1			
CR6 1/	P2-26	P1-11			
CR7	P2-25	P1-1			
CR8	P2-22	P1-11	P2-23	P1-6	Energize K14
CR9	P2-33	P2-21			
CR10 2/	P2-19	P2-32			
CR11 2/	P2-24	P2-32			
CR12 1/	P2-26	P1-6	P2-23	P1-6	Energize K14

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TABLE II. Diode circuits - Continued.

Diode Circuit	Terminals		18 to 30 V dc		Remarks
	Anode	Cathode	Positive	Return	
CR15	P2-25	P2-21	P2-34	P2-20	Energize K3
CR18 2/	P1-6	P2-18	P1-16	P1-6	Energize K20
CR21 1/	P2-14	P1-6	P2-17	P1-6	Energize K15
CR23 1/	P2-14	P1-11			
CR27 1/	P2-2	P1-6	P2-6	P1-6	Energize K16
CR28 1/	P2-2	P1-1			
CR32	P1-35	P1-1			
CR33	P1-35	P2-1			
CR34	P1-30	P1-31			
CR36	P1-27	P2-18	P2-34	P2-16	Energize K4
CR39	P1-13	P1-11			
CR41	P1-24	P1-1			
CR42	P1-10	P1-12			
CR43	P1-10	P2-18	P2-34	P2-16	Energize K4
CR45	P1-8	P1-11			
CR47 1/	P1-3	P1-6	P1-5	P1-6	Energize K12
CR48 1/	P1-2	P1-6	P1-5	P1-6	Energize K12
CR49	P1-3	P1-1			
CR50	P1-2	P1-11			
CR51	P1-21	P1-1			
CR53 2/	P1-6	P1-14	P1-16	P1-6	Energize K20

1/ Forward voltage test only.
2/ Reverse leakage test only.

3.5.4 Blocking diodes. Not applicable.

3.5.5 Resistors. Not applicable.

3.5.6 Relay operation. Relays shall energize when 18 to 30 V dc is applied to the terminals specified in tables III and IV. With 18 to 30 V dc applied to the relay contact terminals specified in tables III and IV and with a 560 ± 56 ohm resistor connected between the relay contact return terminal and the relay contact power supply return terminal, the voltage drop across the resistor shall be as specified in tables III and IV. Unless otherwise specified herein, VH shall be within 0.5 V dc of the contact supply voltage and VL shall not exceed 0.5 V dc. The voltage sequences of table IV, column 6 are defined for uninterrupted application of power to the terminals specified in columns 3, and 5 except where changes are indicated as the sequence progresses (see 4.8.4.7).

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TABLE III. Relay operation.

Relay Circuit	Relay coil		Relay contacts		18-30 V dc		Resistor voltage	
	Positive	Return	Positive	Return	Positive	Return	Energized	De-energized
K1, K2	P2-34	P2-33	P2-21	P1-6			VH	VL
			P2-30	P1-6			VH	VL
			P2-29	P1-6			VH	VL
K3	P2-34	P2-20	P2-33	P2-21			VL	VH 1/
K4	P2-34	P2-16	P1-10	P2-18			VH 1/	VL
			P2-15	P1-6			VH	VL
K5	P2-12	P2-35	P2-13	P2-11			VH	VL
K8	P1-30	P1-6	P2-28	P2-27	P1-29		VH	VL
			P2-28	P2-27	P1-28		VL	VH
			P1-34	P1-32			VH	VL
			P1-34	P1-33			VL	VH
K9	P1-23	P1-25	P1-12	P1-27			VH	VL
			P1-22	P1-27			VL	VH
K10	P1-23	P1-26	P1-13	P1-6			VH	VL
			P1-31	P1-27			VH	VL
K11	P1-9	P1-6	P1-8	P1-6			VH	VL
			P1-31	P1-10			VH	VL
K12	P1-5	P1-6	P1-2	P1-6			VH 1/	VL
			P1-3	P1-6			VH 1/	VL
			P2-30	P1-6	P2-34	P2-33	VL	VH
			P1-7	P1-4			VL	VH
K13	P1-28	P1-6	P2-28	P2-27			VH	VL
K17	P2-6	P1-6	P2-4	P2-8			VH	VL
K18	P1-16	P1-6	P2-13	P2-11			VH	VL
K19	P1-16	P1-6	P2-10	P2-35		P2-1	VH	VL
K20	P1-15	P1-6	P1-25	P1-6			VH	VL
			P2-18	P1-14			VL	VH
			P2-30	P1-6	P2-34	P2-33	VL	VH
			P1-21	P1-6			VH	VL
K21	P1-16	P1-6	P2-22	P1-6	P2-23 & P1-15		VH	VL

1/ VH denotes contact supply voltage (+0, -1.8) V dc.

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TABLE IV. Relay sequential logic.

1 Step	2 Relay circuit	3 Relay coil Positive	3 Relay coil Return	4 Relay contacts Positive	4 Relay contacts Return	5 18-30 V dc Positive	5 18-30 V dc Return	6 Resistor voltage
1	K6, K7	P2-13	P1-6	P2-10	P2-35			VL
2							P2-1	VH
3								VH
4							P2-3	VL
5								VL
1	K13, K14, K15	P2-23	P1-6	P2-22	P1-6			VL
2						P1-28		VL
3								VL
4						P2-17		VL
5						P2-17 & P1-28		VL
6						P1-28		VH
1	K13, K14	P2-23	P1-6	P2-24	P1-6			VL
2		P2-23	P1-6			P1-28		VH
3			P1-6			P1-28		VL
1	K13, K15, K16	P2-17	P1-6	P2-19	P1-6			VL
2		P2-17	P1-6			P2-6		VL
3		P2-17	P1-6			P1-28 & P2-6		VH
4		P2-17	P1-6			P1-28		VL
5		P2-17	P1-6					VL
6		P2-17	P1-6			P2-6		VL
7		P2-17	P1-6			P1-28 & P2-6		VH
8			P1-6			P1-28 & P2-6		VL
1	K2, K16	P2-6	P1-6	P2-29	P1-6			VL
2		P2-6	P1-6			P2-34	P2-33	VL
3			P1-6			P2-34	P2-33	VH

3.5.7 Transient voltage characteristics.

3.5.7.1 Internal relays. Upon removal of 18 to 30 V dc from the terminals specified in table V, the relay coil dropout transient voltage shall be within the range of ± 32 V dc unless otherwise specified (see 4.8.4.8.1).

3.5.7.2 External relays. Unless otherwise specified, with 18 to 30 V dc ied across relay coils conforming to table VI and connected as specified therein, the dropout transient voltage shall be within the range of ± 32 V dc upon removal of the applied voltage (see 4.8.4.8.2).

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TABLE V. Transient suppression, internal relays.

Relay Circuit	Relay Coil Positive	Relay Coil Return	Suppression Components	Remarks
K1, K2				See table VI.
K3	P2-34	P2-20	CR2, VR2	
K4	P2-34	P2-16	CR3, VR2	
K5	P2-12	P2-35	CR24, VR3	
K6	P2-13	P2-1	CR29, VR1	
K7	P2-13	P2-3	CR30, VR1	
K8	P1-30	P1-6	CR35, VR8	
K9	P1-23	P1-25	CR38, VR9	
K10	P1-23	P1-26	CR37, VR9	
K11	P1-9	P1-6	CR44, VR8	
K12	P1-5	P1-6	CR46, VR8	
K13	P1-28	P1-6	CR40, VR8	
K14	P2-23	P1-6	CR14, VR5	
K15	P2-17	P1-6	CR19, VR6	
K16, K17	P2-6	P1-6	CR26, VR7	
K18	P1-16	P1-6	CR52, VR8	
K19, K20, K21	P1-15	P1-6	CR54, VR8	Energize K18

TABLE VI. Transient suppression, external relays.

External Relay	Relay Coil Positive	Relay Coil Return	Suppression Components	Remarks
K1, K7	P2-34	P2-33	CR1, VR2	4 relays in parallel: K1, K7, A4K1, A4K2
				K1: MS24171-D1 per MS24171 K7: M6106/19-011 per MIL-R-6106/19
K2	P2-32	P2-19	CR10, VR4	MS24187-D1 per MS24187
K4	P2-18	P1-6	CR18, VR8	M5757/23-001 per MIL-R-5757/23 Energize K18 and K20. Apply 18 to 30 V dc to P1-15, P1-16 (positive) and P1-6 (return)
K5	P1-14	P1-6	CR53, VR8	See K4 note
K11	P2-32	P2-24	CR11, VR4	M6106/19-011 per MIL-R-6106/19

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3.5.8 Time delay and latching circuit. Not applicable

3.5.9 Fire warning circuit. Not applicable.

4.8 Methods of inspection.

4.8.3 Weight. Conformance to 3.4.2 shall be determined by weighing the PCC (TNB-A4) on a scale.

4.8.4.3 Circuit isolation. To determine conformance to 3.5.2, the PCC shall be tested in accordance with MIL-STD-202, method 302, test condition A. Connect the terminals listed in table I for each circuit number and measure the circuit resistance from each circuit to all of the remaining circuits.

4.8.4.4 Diode I/V characteristics.

4.8.4.4.1 Forward voltage drop. To determine conformance to 3.5.3.1, the forward voltage drop of the diodes specified in table II shall be measured in accordance with MIL-STD-750, method 4011.4. When energization of a relay is required, the relay associated with the diode by applying 24 ± 2 V dc between the relay terminals specified in table II using a second power source whose power return terminal is electrically isolated from the primary power source used for testing the diodes.

4.8,4.4.2 Reverse leakage current. To determine conformance to 3.5.3.2, the reverse leakage current of the diodes specified in table II shall be measured in accordance with MIL-STD-750, method 4016.3. When energization of relay is required, energize the relay associated with the diode by applying 24 ± 2 V dc between the relay terminals specified in table II using a second power source whose power return terminal is electrically isolated from the primary power source used for testing the diodes.

4.8.4.5 Blocking diode. Not applicable.

4.8.4.6 Resistors. Not applicable.

4.8.4.7 Relay operation. TO determine conformance to 3.5.6, connect the positive terminal of a power source set to 18.0 (+0.5, -0.0) V dc to each of the relay contact positive terminals of the relay under test in tables III and IV. Connect a 560 ± 56 ohm, 2 watt minimum resistor between the relay contact return terminal(s) and the return terminal of the power source. Connect a second power source, whose power return terminal is electrically isolated from the previous power source, between each of the relay coil terminals of tables III and IV and (when required) between any additional terminals of tables II and III that require 18 to 30 V dc. With power applied, the voltage drop across each of the resistors shall be measured to conform to 3.5.6, table III and IV. The relays shall de-energized by removing the voltage from the relay coil positive terminal and the Voltage drop across the resistors shall be measured to conform to 3.5.6, table III and IV. Tests on the relays of table IV must be performed in step sequence shown. The entire test shall be repeated except the power source voltagers shall be set to 30 (+0.0, -0.5) V dc.

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4.8.4.8 Transient voltage characteristics.

4.8.4.8.1 Internal relays. To determine conformance to 3.5.7.1, apply 30.0 (+0.0, -0.5) V dc and connect a device for measuring voltage transients between the terminals of each relay specified in table V. Remove the applied voltage from the positive terminal and verify that the voltage transient measured conforms to the requirement of 3.5.7.1.

4.8.4.8.2 External relays. To determine conformance to 3.5.7.2, connect the coil of a relay conforming to table VI and a device for measuring voltage transients between each of the relay coil terminals specified in table VI and apply 30.0 (+0.0, -0.05) V dc to the terminals. Remove the voltage from the positive terminal and verify that the voltage transient measured conforms to the requirements of 3.5.7.2.

4.8.4.9 Time delay and latching circuit. Not applicable.

4.8.4.10 Fire warning circuit. Not applicable.

4.8.4.11.2 Basic shock. The test for basic shock shall be as specified in the general specification except for intermittent operation. Intermittent operation shall be tested as follows: monitor the relay contacts listed in table VII for contact chatter or false make. The voltage applied to energize the relay coils shall be 18.0 (+0.5, -0.0) V dc.

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